Pranjal Kalekar

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Summary

Applied Data Scientist with a proven history of spearheading cross-functional data initiatives from initial prototype to global scale. Dedicated to delivering secure, high-quality AI and data products that drive measurable growth and operational excellence.

TECHNICAL SKILLS

- **Tech Languages**: Python, MySQL, MS SQL, R, MATLAB, No-SQL, C++, HTML/CSS
- **Databases**: MongoDB, MariaDB, Neo4j, Apache Spark, PostgreSQL, NoSQL, RDBMS
- **Tools**: Git, VS code, PyCharm, Google Cloud Platform, Tableau, PowerBI, WordPress, Azure, Hadoop
- **Libraries**: PyTorch, Plotly, Tensorflow, SciPy, pandas, Scikit-learn, dlib, seaborn, NLTK, XGBoost, NetworkX
- **Platforms**: Microsoft Excel, Google Colab, PySpark, MS Office, Unix/Linux, MySQLWorkbench, R Studio

EDUCATION

Northeastern University **Sep 2022 - Dec 2024**
Master of Science, Data Analytics Engineering *Boston, MA*

VJTI **Aug 2018 - May 2022**
B.Tech, Electronics Engineering *Mumbai, MH, India*

EXPERIENCE

Haas Lab, Mass General Hospital- Harvard Medical School **Mar 2025 - Present**
Research Engineer *Boston, MA*

- Engineered a graph-based imputation pipeline using machine learning, reducing data sparsity and achieving 35% better performance.
- Deployed GPU-optimized XGBoost and Neural Networks, enhancing accuracy by 40% over linear models and isolating critical biological feature importance.
- Stabilized GNN training by refactoring the architecture and applying systematic hyperparameter tuning in PyTorch, surpassing benchmark performance by 25% and improving model reliability

DataZ **Jan 2025 - Mar 2025**
Associate *Remote - Boston, MA*

- Built Python-based web scraping pipelines using Selenium, Scrapy, BeautifulSoup, and ScraperAPI to collect product data from sites like Alibaba and AliExpress, and stored 30K+ entries in a MySQL database, enabling structured analytics and automated reporting
- Delivered actionable insights and Tableau dashboards tracking key engagement metrics, supporting investor updates and product decisions in a lean early startup environment.

Haas Lab, Mass General Hospital- Harvard Medical School **Jan 2024 - Sep 2024**
Data Analyst-1 *Boston, MA*

- Analyzed breast cancer proteomics data from mass spectrometers using clustering (K-means, hierarchical, DBSCAN), dimensionality reduction (UMAP, PCA, t-SNE), and imputation (Combat, LUPINE) to improve data quality and clarify tumor subtypes
- Built a graph database from 302 proteomics datasets and trained GNN models, achieving up to 96% AUC-ROC in identifying disease-related protein interactions.
- Trained a node-classification model that achieved a maximum 96% AUC-ROC for individual graphs. Created posters and presented findings in research gatherings to support cross-lab collaborations

Saint Louis University **Oct 2021 - Nov 2021**
Data Visualization Intern *Remote*

- Performed profit focused Meta ads analysis on user performance data, resulting in a monthly subscriber increase of 5% on the client's Meta page.
- Developed NLP-based features through sentiment analysis of user comments using Python and pandas, enabling the team to identify key sentiment trends and improve content targeting
- Built interactive data visualizations with Tableau, Seaborn, and Matplotlib, allowing stakeholders to make informed strategic business decisions

Eddy Tools and Tech Solutions **Jun 2021 - Sep 2021**
Computer Vision Intern *Remote*

- Worked as Project head for an exclusive POC project, 'live attention tracking system' and successfully delivered the project as per the clients requirements. Worked as a project head for 11 teammates and tech team of Eddytools and tech solutions.
- Developed and presented a live attention tracking system using facial landmarks and variance of live transmission or recording for proctoring client within a 3-month span
- Utilized Agile methodology with Jira to monitor and enhance the daily productivity of a team of 5, achieving a 10% performance boost, demonstrating expertise in project management.

• Built interactive data visualizations utilizing Tableau, pandas, and Power BI to support strategic business decisions, showcasing proficiency in analytics, BI, statistics, and data science.

AIRPIX **Jun 2020 - Sep 2020**

AI, ML AND EDGE COMPUTING INTERN *Mumbai*

- Created a large and well-labeled dataset using Python and YOLO object detection algorithm; effective on low-quality CCTV images.
- Trained a TensorFlow model to detect Personal Protective Equipment (PPE) in high-risk industrial and mining sites, delivering reliable detection that supports safety monitoring
- Collaborated with the team to optimize the training and inference pipelines, resulting in faster and more accurate results and improved problem solving, handling data structure and debugging skills.

PROJECTS

KLCE & GIB Regularization - Class Imbalance Resolution **Sep 2023 - Dec 2023**

- Spearheaded an independent research project to enhance Graph Neural Network (GNN) node classification accuracy by addressing pervasive class imbalance issues.
- Implemented novel KLCE (Kullback-Leibler Cross-Entropy) regularization loss (ICLR 2024), achieving performance improvements ranging from 0.9% to 29% over traditional log loss and exceeding published results through rigorous Optuna hyperparameter tuning.
- Improved balanced accuracy by 10% on the PubMed dataset using Graph Information Bottleneck (GIB)(NeurIPS 2020) regularization based loss function, optimizing node representations by balancing relevance and compression.
- Utilized PyTorch and PyTorch Geometric for GNN model implementation and custom loss function development, evaluating performance with F1-score, Balanced Accuracy, Precision, Recall, and AUC-ROC.

CredRisk **Sep 2023 - Dec 2023**

- A Credit Risk forecasting Predictor Model
- Ran ETL process using Python on LendingClub data for extraction, cleaned, processed, and fine-tuned the data for future extractions with correlation with the target variable.
- Visualized data to get the insights using Tableau dashboard and Python libraries.
- executed different machine learning algorithms such as XGB classifier, Random Forest Tree, Logistic Regression with SMOTE oversampling, etc. reaching a maximum balanced accuracy of 89% with a specificity score of 0.97 for the class of interest i.e. high-risk loans

All-in-one Airport Management System **May 2022**

- Using ETL pipelining Created an RDBMS substantially containing the entities in Airport Management system including terminals, pilots, employees, stores, etc. from different data sources. created/ connected the entities with specific access control
- Got insights into the system using CTEs, window functions, and different MySQL Queries.
- Visualized the database using Tableau dashboard and Python Libraries Such as Matplotlib, Plotly, and Seaborn

Fuel-IT-Right **Nov 2025**

Sundai

- Live Calorie burn calculator for different intensity level workouts using different sensors on an athlete.
- Encapsulated in a apple watch app and a webpage connected to heart rate and power sensors using Swift Studio.

Satellite Imagery Gap Filling using CNN **Sep 2025**

Wood-well Climate research center

- Build a CNN Model to predict the gaps in the Landsat dataset for further processing trained on AlphaEarth and modis dataset.
- Ran ETL process using Python on LendingClub data for extraction, cleaned, processed, and fine-tuned the data for future extractions with correlation with the target variable.

LLM based OS agent for iOS **May 2022**

Sundai

- Developed a stateless AI Agent orchestration layer that integrates LLMs (GPT-4o) with native macOS UI automation to perform real-time task execution.
- Reduced pipeline latency by optimizing core script execution and implementing efficient data handling techniques.
- Leveraged native macOS Accessibility permissions (AXIsProcessTrusted), ensuring full compliance with system-level security protocols and explicit user consent.

CERTIFICATES AND LEADERSHIP

- Hacker at Sundai MIT- Harvard AI Prototype Launching Club, Kaggle Advanced SQL certificate, Former VP of Relations and Projects Data Science Hub, Graduate Certification in Leadership with Northeastern University.
- Mentor at Wellmet transitional housing, MIT- ESP splash Teacher 2025, NU cycling- Road Biking ECCC Racing, WRBB Radio Show Host, Climate Justice Fellow 2023, Fiber artist - Pranpaan, MIT- ESP splash Teacher.